

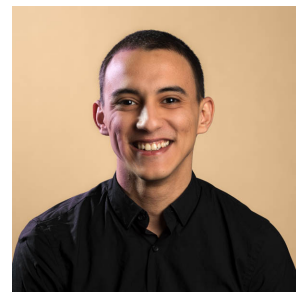
Yacine Belal, PhD Candidate

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🌐 Yacine Belal

📍 Lyon, France



Education

- Jan. 2022 – June 2025 📖 **Ph.D. in Computer Science** — INSA Lyon & iExec Blockchain Tech
Thesis title: *Collaborative Learning: Personalization, Privacy, and Robustness at the Edge*
Advisor: Dr. Sonia Ben Mokhtar
Expected defense: June 10, 2025
- Sep. 2019 – Sep. 2021 📖 **M.Sc. in Computer Science** — University of Western Brittany
Specialization: *Intelligent, Interactive, and Autonomous Systems*

Publications

Journal Articles

- 1 Y. Belal, S. Ben Mokhtar, H. Haddadi, J. Wang, and A. Mashhadi, "Survey of federated learning models for spatial-temporal mobility applications," *ACM Transactions on Spatial Algorithms and Systems*, vol. 10, no. 3, pp. 1–39, 2024. 🔗 DOI: 10.1145/3666089.
- 2 Y. Belal, A. Bellet, S. B. Mokhtar, and V. Nitu, "PEPPER: empowering user-centric recommender systems over gossip learning," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 6, no. 3, 101:1–101:27, 2022. 🔗 DOI: 10.1145/3550302.

Conference Proceedings

- 1 Y. Belal, M. Maouche, S. B. Mokhtar, and A. Simonet-Boulogne, "Inferring communities of interest in collaborative learning-based recommender systems," in *Proceedings of the IEEE International Conference on Distributed Computing Systems (ICDCS)*, To appear, 2025. 🔗 URL: <https://hal.science/hal-05007813>.
- 2 S. Yadav, Y. Belal, B. Lagesse, and A. Mashhadi, "Benchmarking clustered federated learning algorithms for next-point prediction," in *Proceedings of the IEEE International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, To appear, 2025.

Pre-Prints

- 1 O. Touat, J. Brunon, Y. Belal, *et al.*, *Scrutinizing the vulnerability of decentralized learning to membership inference attacks*, 2024.

Research Experience

- Oct. 2022 – June 2024 📖 **Intern Co-Supervisor** — INSA Lyon
- Supervised two Master's students on decentralized anomaly detection in ECG signals.
 - Supervised one Bachelor's student on adversarial attacks in federated learning.
- Jan. 2021 – Aug. 2021 📖 **Research Intern** — LIRIS Lab, INSA Lyon
- Implemented gossip learning algorithms for recommendation and sentiment analysis.
 - Reviewed the literature on personalized federated learning.

Teaching Experience

- 2022–2024 📖 **Linear Algebra**
- Delivered 48 hours of lectures for first-year engineering students.

Teaching Experience (continued)

- 2022–2023  **Privacy-Preserving Distributed Machine Learning**
Delivered 24 hours of lab sessions to fifth-year engineering students.
Covered: distributed architectures, federated learning, differential privacy, and data heterogeneity.
- 2021–2022  **Cloud Computing and Big Data Applications**
Delivered 40 hours of lab sessions to fifth-year engineering students.
Covered: web scraping, data storage and processing in AWS, and data analysis/visualization with pandas.

Technical Skills

- | | |
|--------------------------------|---|
| Programming |  Python, C++, CUDA, Java. |
| Machine learning Frameworks |  Pytorch, Scikit-Learn. |
| Privacy Enhancing Technologies |  Differential Privacy (Octopus, DP-SGD). |
| Collaboration tools |  Git, Docker, Weights & Biases. |

Miscellaneous

Notable Talks

- Jan. 2025  **PEPR Cybersécurité Winter School** — Autrans, France
Title: *Byzantine Resilience in Dynamic and Sparse Gossip Learning*
- Mar. 2024  **PEPR IA Workshop** — Grenoble, France
Title: *Byzantine Resilience in Gossip Learning*
- Apr. 2023  **Atelier sur la Protection de la Vie Privée APVP 2023** — Besançon, France
Title: *Inferring Communities of Interest in Collaborative Learning-based Recommender Systems*
- Sept. 2022  **IMWUT / UbiComp 2022** — Cambridge, UK
Title: *PEPPER: Empowering User-Centric Recommender Systems over Gossip Learning*

Academic Service

-  Peer-reviewer for Information Sciences, 2023.
-  Student Volunteer, The Web Conference 2023.

Certificates

-  **Convex Optimization** - Imperial College London
-  **Statistics with R** - Université Paris-Saclay
-  **Getting Started with Accelerated Computing in Cuda C/C++** - NVIDIA

Languages

-  Arabic (native), French (fluent), English (fluent)

References

Available on Request